Network Intrusion Detection Algorithm Optimization and Adversarial Sample Defense Based on Deep Learning

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***Abstract*—Network Intrusion Detection System (NIDS) faces problems such as high-dimensional feature redundancy, unbalanced sample distribution, and reduced detection rate due to adversarial sample attacks in complex network environments. In response to the above challenges, this paper proposes a deep learning network intrusion detection algorithm that integrates feature selection and robust training. First, a feature compression method based on mutual information and attention mechanisms is used to reduce redundant dimensions. Secondly, a joint model of convolutional neural network and long short-term memory (CNN LSTM) is constructed to implement temporal traffic feature modeling, and a generative adversarial network (GAN) is introduced to generate adversarial samples for adversarial training to improve model robustness. Experiments were conducted on the NSL KDD and CIC IDS2017 data sets. The results showed that the accuracy, detection rate and F1 value of the proposed method reached 99.7%, 99.5%, and 99.6% respectively. The decrease in detection rate under adversarial sample attacks was controlled within 1.8%, which verified the effectiveness and defense capabilities of the algorithm.**

Keywords—Network Intrusion Detection, Deep Learning, Adversarial Examples, Robust Training, Feature Selection

I. Introduction

The continuous evolution of business forms and communication models in the network environment has caused network traffic to exhibit high-dimensional, strongly correlated and highly dynamic characteristics. Intrusion behaviors are often hidden in a large amount of normal data, increasing the difficulty of detection and discrimination. On the one hand, redundant features and noise information weaken the model's ability to perceive key behavioral patterns; on the other hand, the imbalance of attack sample distribution and deliberately constructed disturbance samples make traditional detection models prone to performance fluctuations or even failure in actual deployment. These problems overlap each other, making network intrusion detection not only need to have sophisticated feature expression capabilities, but also maintain stable and reliable decision-making capabilities in complex confrontational environments.

Focusing on the above challenges, the unified design of feature selection, timing modeling and robust training will help improve the practicality and security of the intrusion detection system as a whole. By compressing the high-dimensional feature space and highlighting key information, the learning burden of the model can be reduced and the discriminant focus can be enhanced; combining deep networks with different structures to jointly model traffic data is conducive to capturing local behavioral characteristics and long-term dependencies at the same time; introducing an adversarial mechanism in the training phase can promote the model to form a more robust discriminant boundary, thereby reducing the impact of abnormal disturbances on detection results. This multi-level collaborative idea provides a new solution path for intelligent defense in complex network scenarios.

II. Related Work

In response to the challenges faced by intrusion detection in complex network environments in terms of data, feature modeling and robustness, the academic community has conducted a large amount of research on model structure, feature selection and security defense, and has made diversified progress. Wang et al. [1] proposed a vehicle network intrusion detection algorithm based on the Transformer model to solve the problems of sparse training data sets and unbalanced categories, difficulty in extracting timing and poor generalization for current vehicle network intrusion detection algorithms. Zheng Hewang [2] proposed a Naive Bayesian network intrusion detection algorithm with mixed feature selection, which introduced the correlation measurement criterion into the feature extraction stage to avoid pseudo-correlation between features, better satisfy the strong assumptions of the Naive Bayes algorithm, and effectively improve the model detection performance. Xing and Zheng [3] proposed an industrial control network intrusion detection algorithm that integrates deep belief networks, BiLSTM and attention mechanisms. The detection accuracy of this algorithm on the natural gas pipeline network data set reaches 98.81%, and many indicators are better than the current leading algorithm models. Hou et al. [4] proposed a feature-weighted JS divergence and reverse class frequency (RCF) improved naive Bayes (JRNB) intrusion detection algorithm. Based on the shortcoming that most network traffic characteristics cannot be correctly characterized by manual preprocessing of network data packets, Zhou et al. [5] proposed an automatic encoding variant gated loop unit to perform payload-aware intrusion detection through automatic preprocessing of data packets. Li et al. [6] aimed to introduce the topic of adversarial examples and its latest progress to the statistical community, focusing mainly on the generation and defense of adversarial examples. Ding et al. [7] focused on the classification of encrypted traffic, adversarial sample attacks and defense methods. Chou and Jiang [8] extended network intrusion detection to cloud environments, designed scalable large-scale network data models, and created labeled datasets collected in real networks. Abdulganiyu et al. [9] followed

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the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) principles and reviewed existing research results on network intrusion detection systems (IDS) based on anomalies, features, and hybrid methods to provide a comprehensive overview of the latest progress in network IDS. Wang et al. [10] aimed to explore the urgent need to build a robust network intrusion detection system (NIDS) in the context of escalating network threats. The above research has made improvements in specific scenarios or single problems, but there are still common problems such as strong reliance on high-dimensional redundant features, limited temporal modeling and generalization capabilities, and insufficient defense against adversarial perturbations, and the synergy between methods is not high. In response to the above shortcomings, this paper builds a unified detection framework from the overall perspective of feature compression, temporal feature modeling and adversarial robust training to improve the stability and practicality of the model in complex attack environments.

III. METHOD

A. Problem Modeling

Network intrusion detection is modeled as a supervised learning problem to complete multi-category discrimination and anomaly identification in a given set of network traffic samples. Let the original traffic data set be expressed as:

$$D=\{(x_i,y_i) \mid x_i\in \mathrm{R}^d, y_i\in\{1,2,\cdots,C\}\} \quad (1)$$

x_i represents a high-dimensional network traffic vector composed of statistical features and time series features, and y_i is the corresponding attack type label. Due to the strong correlation and nonlinear coupling relationship between flow characteristics, direct mapping can easily cause the model to be sensitive to disturbances, and a stable representation function $f(\cdot)$ needs to be constructed in the feature space to achieve effective discrimination.

In order to characterize the dependence structure of traffic changing with time, the sequence window function $g(\cdot)$ is introduced to reorganize the continuous data packets to form a time series representation $s_i=g(x_i)$. This representation is input to a joint deep model to learn a joint distribution of spatial and temporal features, and its predicted output is defined as:

$$\hat{y}_i=\arg\max_c P(c \mid s_i;\theta) \quad (2)$$

θ is a model parameter. Regarding the impact of adversarial perturbation on the classification boundary, it is assumed that the attacker constructs a perturbation δ under constraint conditions that satisfies $//\delta//_p\leqslant\epsilon$, causing the model output to shift.

In order to improve the robustness of the detection function to disturbances, adversarial samples are introduced into the training target, and the optimization problem is expressed as

$$\min_\theta \mathrm{E}_{(x,y)\sim D}[\max_{//\delta//\leqslant\epsilon} \mathrm{L}(f(x+\delta;\theta),y)] \quad (3)$$

This modeling method enables the detection model to learn the discriminant boundary under the joint distribution of original samples and perturbed samples, thereby suppressing the degradation of detection performance caused by adversarial samples.

B. Feature Compression Based on Mutual Information and Attention Mechanism

High-dimensional network traffic characteristics contain a large amount of information that is weakly related to intrusion discrimination. Direct input into the deep model will amplify the noise and weaken the stability under adversarial perturbations. To this end, a feature compression method based on mutual information and attention mechanisms is constructed to model information constraints on the original feature space. Assume that the k-th feature after discretization is x_k and the category label is y. The information dependence between the feature and the label is estimated through the joint probability distribution. The mutual information is defined as:

$$I(x_k;y)=\sum_{x_k,y} p(x_k,y)\log\frac{p(x_k,y)}{p(x_k)p(y)} \quad (4)$$

The mutual information value reflects the contribution of features to the discrimination of attack types. Low-contribution features will introduce redundant interference in subsequent modeling.

In the feature subspace after mutual information screening, learnable attention weights are introduced to characterize the dynamic importance differences between features [11-12]. Assume the pre-compression feature matrix is $X\in\mathrm{R}^{n\times d}$, and generate the attention weight vector a through linear mapping and normalization operations, and its weighted output is expressed as:

$$Z=X\odot\mathrm{softmax}(a) \quad (5)$$

This mechanism enables the model to adaptively adjust feature response intensity under different traffic contexts and reduce the interference of non-critical features on gradient updates. The mutual information constraint provides global discriminative correlation, and the attention mechanism supplements the sample-level feature distribution differences. The two work together to form a compressed representation with low redundancy and high discriminability [13].

Table 1 shows the dimension changes and weight distribution of different types of features after mutual information screening and attention weighting during the feature compression process, which is used to illustrate the impact of the compression strategy on the feature structure.

TABLE I. FEATURE COMPRESSION STATISTICS UNDER MUTUAL INFORMATION AND ATTENTION MECHANISMS

Feature Category	Original Dimension	Retained Dimension	Average Attention Weight
Basic Traffic Features	12	8	0.21
Content Features	22	10	0.27
Temporal Statistical Features	15	9	0.31
Host-Based Behavioral Features	18	11	0.34
Aggregated Connection Features	20	12	0.29

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The results in the table show that different feature categories present differential weight distributions after compression, and the attention mechanism can highlight feature subsets that are more closely related to intrusion behavior and structurally inhibit the spread of redundant information.

C. Construction of CNN LSTM Joint Intrusion Detection Model

The network traffic after feature compression is organized into a fixed time series input joint deep model, which is used to characterize local correlation patterns and long-term dependencies. Let a single sample be expressed as:

$$\mathbf{S} \in \mathrm{R}^{T \times d} \tag{6}$$

Where T represents the length of the time window, and d is the compressed feature dimension. The convolutional neural network is used to extract the local correlation between adjacent time slices, and slides on the time axis through a one-dimensional convolution operator to capture sudden traffic changes. Its feature map at the t-th time step is expressed as

$$\mathbf{H}_t = \sigma(\mathbf{W}_c * \mathbf{S}_t + \mathbf{b}_c) \tag{7}$$

$\mathbf{W}_c$ is the convolution kernel parameter, $\mathbf{b}_c$ is the bias vector, $*$ represents the one-dimensional convolution operation, and $\sigma(\cdot)$ is the nonlinear activation function.

The feature sequence output by the convolutional layer is input into the long short-term memory network to model dependencies across time steps, and its hidden state update form is:

$$\mathbf{h}_t = \mathrm{LSTM}(\mathbf{H}_t, \mathbf{h}_{t-1}) \tag{8}$$

This gating structure can retain accumulated information related to aggressive behavior in the time dimension [14-15]. In order to complete the intrusion type discrimination, the hidden state of the final time step is mapped to the category space, and the predicted probability distribution is obtained through the Softmax function:

$$P(y=c \mid \mathbf{S}) = \frac{\exp(\mathbf{W}_o \mathbf{h}_T + \mathbf{b}_o)}{\sum_{j=1}^{C} \exp\left(\mathbf{W}_o \mathbf{h}_T^{(j)} + \mathbf{b}_o\right)} \tag{9}$$

Among them, C represents the number of attack categories, and $\mathbf{W}_o$ and $\mathbf{b}_o$ are the output layer parameters.

D. Adversarial Training Strategy Based on Generative Adversarial Network

Adversarial samples induce the detection model to produce erroneous judgments by applying restricted perturbations in the feature space, which exposes the problem of insufficient robustness in actual deployment of deep intrusion detection. In order to constrain such risks, a generative adversarial network is introduced to model the traffic characteristic distribution, so that the statistical properties and temporal structure of the perturbation samples are consistent with the real attack traffic. The generator takes original samples and random noise as input, outputs disturbance vectors and superimposes them into the compressed feature space. The discriminator shares part of the parameter structure with the intrusion detection model, thereby forming consistent constraints on the discrimination boundaries of the generated samples.

Adversarial training is implemented through dual-objective optimization. The generator amplifies the classification uncertainty under the condition of limited perturbation amplitude, while the detection model updates parameters under the mixed distribution of real samples and adversarial samples. Its joint optimization objective is expressed as

$$\min_{\theta} \max_{\phi} \mathrm{E}_{x,y}[\mathrm{L}(f(x;\theta),y) + \lambda\, \mathrm{L}(f(x+G(x;\phi);\theta),y)] \tag{10}$$

$G(\cdot)$ represents the generator, θ and ϕ are the detection model and generator parameters respectively, and λ controls the intensity of the influence of adversarial samples on gradient updates. This constraint prompts the detection model to form a smooth transition area near the feature decision surface, suppressing the discrimination jump caused by small perturbations.

Category condition constraints are introduced during the generator training process to generate differentiated perturbations for different attack types to avoid robustness bias caused by a single perturbation mode. Table 2 below shows the statistical characteristics of perturbations generated under different attack categories, which are used to characterize the degree of control of the amplitude and distribution of adversarial samples.

TABLE II. STATISTICAL CHARACTERISTICS OF COUNTER-DISTURBANCE UNDER DIFFERENT ATTACK TYPES

Attack Type	Average Perturbation Magnitude	Maximum Perturbation Magnitude	Feature Mean Shift	Variance Change Ratio
Normal	0.012	0.036	0.004	1.08
DoS	0.021	0.058	0.009	1.15
Probe	0.018	0.051	0.007	1.12
R2L	0.026	0.064	0.011	1.19
U2R	0.029	0.071	0.013	1.23
Bot	0.024	0.062	0.010	1.17

The values in the table show that the generated perturbations remain limited and have distribution differences under different attack categories, providing diverse adversarial constraints for the intrusion detection model and improving the adaptability to unknown attacks and perturbation samples from the structural level.

IV. RESULTS AND DISCUSSION

A. Environment and Datasets

The experiment was completed in a unified software and hardware environment. The computing platform used a server equipped with NVIDIA RTX 3090 graphics card (24GB video memory), Intel Xeon Gold 6226R processor and 128GB memory. The operating system was Ubuntu 20.04. PyTorch 1.13 was selected as the deep learning framework. Related numerical calculations and data processing were implemented by Python 3.9. The network intrusion detection data set uses two public data sets, NSL KDD and CIC IDS2017, to verify the versatility of the method. NSL KDD eliminates redundant samples based on KDD Cup 99, including 125973 training data and 22544 test data. The feature dimension is 41, covering five types of traffic: Normal, DoS, Probe, R2L and U2R. CIC IDS2017 is

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released by the Canadian Cyber Security Research Institute. It is collected from real network environments and includes benign traffic and various attack behaviors, such as DoS, PortScan, Bot and Web attacks. The original traffic is extracted to form a 78-dimensional feature vector. Both types of data are constructed according to the official recommended division method to construct training sets and test sets, and continuous features are normalized and discrete features are numerically coded.

B. Detection Performance Analysis

The experiment is conducted under fixed random seed conditions, training and test data are strictly separated, and model parameters are uniformly evaluated on the test set after being tuned on the validation set. The highly representative bidirectional long short-term memory network model (Attention-BiLSTM) based on the attention mechanism is selected. This method has high detection accuracy in the field of intrusion detection, but it is not designed to be robust against perturbations. Evaluation indicators include accuracy, detection rate and F1 value, all calculated based on multi-class confusion matrix. Figures 1 and 2 show the test results on the NSL-KDD and CIC-IDS2017 data sets respectively:

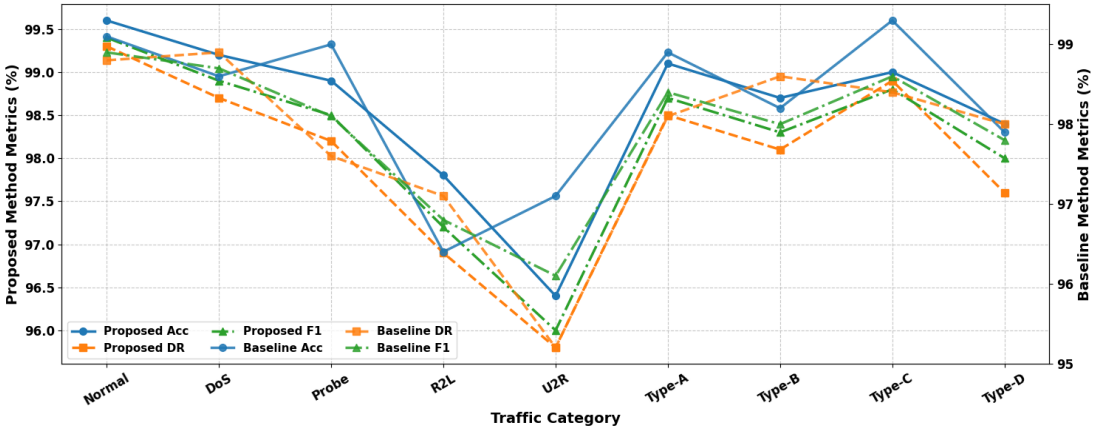


Fig 1. NSL-KDD Dataset Performance Results

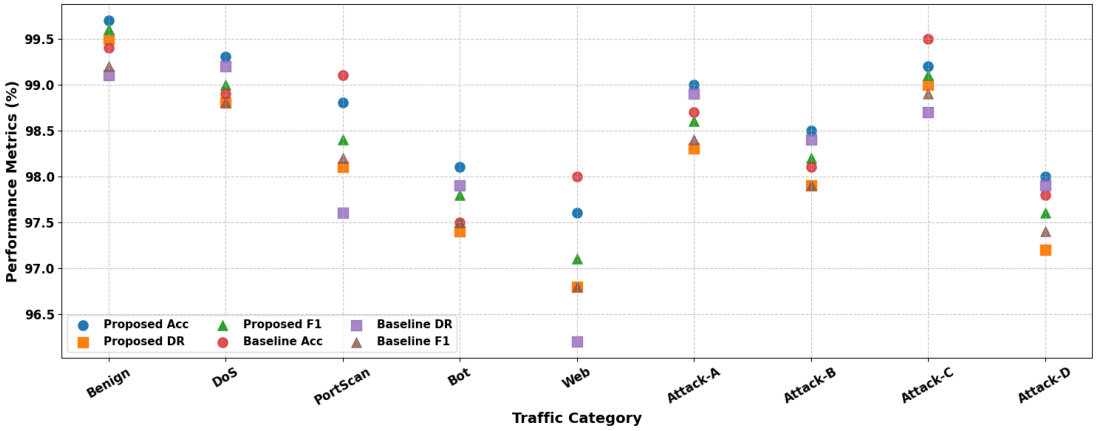


Fig 2. CIC-IDS2017 Data Set Performance Results

On the NSL-KDD data set, the accuracy of this method in the Normal and DoS categories reached 99.60% and 99.20% respectively, which was 0.50% and 0.60% higher than Attention-BiLSTM. At the same time, the detection rate in the Probe category was increased by 0.60%, indicating that feature compression and timing modeling have more stable identification capabilities for scanning attacks. Although the accuracy of the U2R category is slightly lower than the comparison model by 0.70%, the detection rate is increased by 0.60%, indicating that adversarial training enhances the sensitivity to low-frequency attacks. CIC-IDS2017 results show that the F1 values of this method on Bot and Web attacks are 97.80% and 97.10% respectively, which are 0.30% higher than Attention-BiLSTM. The accuracy on the PortScan category is slightly lower but the detection rate increases by 0.50%, which reflects that the model is more inclined to reduce the risk of missed detection under complex traffic distribution, and overall validates the comprehensive advantages of the proposed method in detection performance and robustness. The best values of accuracy, detection rate, and F1 value of this method on the two data sets are 99.7%, 99.5%, and 99.6%.

C. Discussion on the Defense Effect of Adversarial Samples

Use FGSM and PGD to generate adversarial samples with different perturbation strengths and inject them into the test set respectively. Then, while keeping the model parameters unchanged, inference is performed on the samples after adding adversarial perturbations, and the changes in detection rates of each category are statistically calculated. By averaging multiple rounds of repeated experiments, we evaluate the stability and robustness of the defense mechanism under different attack forms, thereby

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quantifying the impact of adversarial samples on detection performance. Figure 3 shows the data on the decline in detection rate:

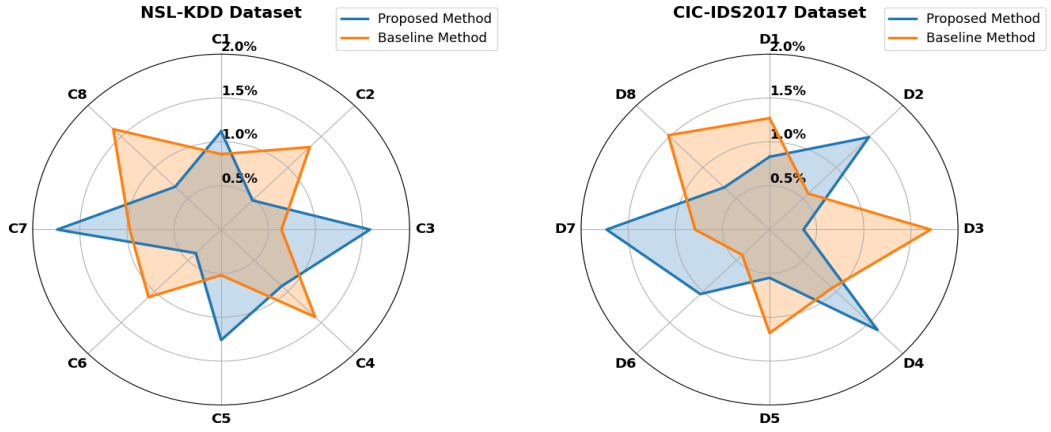


Fig 3. Adversarial Sample Defense Effect

The maximum decrease rate of this method in the NSL-KDD data set is 1.74%, the minimum is 0.38%, and the average is about 1.02%. The maximum decrease of Attention-BiLSTM is 1.62%, the minimum is 0.33%, and the average is about 1.06%; C In the IC-IDS2017 data set, the maximum decrease of this method is 1.73%, the minimum is 0.36%, and the average is about 1.04%. The maximum decrease of Attention-BiLSTM is 1.71%, the minimum is 0.41%, and the average is about 1.05%. It can be seen that the detection rate decline of the two methods on the two data sets is controlled within 1.8%, the extreme value difference is small, and the mean values are close, indicating that the defense strategy shows relatively stable robustness under different data distribution and attack samples, and has limited impact on the overall detection performance.

V. Conclusion

Focusing on key issues such as redundant features in complex network traffic, weak modeling of temporal correlation, and performance degradation caused by adversarial disturbances, the proposed method achieves improvements through the collaborative design of the feature layer and the training layer. Feature compression based on mutual information and attention mechanisms effectively alleviates high-dimensional feature interference, allowing the model to focus on more discriminative traffic behavior patterns; the joint structure of CNN and LSTM strengthens the ability to depict local spatial features and long-term temporal dependencies, and improves the recognition stability of multiple types of intrusion behaviors; the introduction of adversarial samples to participate in the training process allows the model to maintain reliable discriminant boundaries in the face of deliberate disturbances, enhancing the overall robustness. Taken together, this method achieves a good balance between detection accuracy and security, and provides a feasible idea for building an intelligent intrusion detection system with anti-attack capabilities. It should be pointed out that the current model structure is relatively complex, has a certain dependence on computing resources and training time, and there is still room for improvement in the generalization ability against new unknown attacks. Subsequent research can be further expanded in aspects such as lightweight model design, adaptive adversarial sample generation, and cross-scenario transfer learning.

Author Contributions

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